

Linear systems and elimination

Exact extremizers for partial pivoting on orthogonal matrices

Difficulty: challenging **Importance:** interesting to specialist**Status:** [Lean verified](#)

Rating rationale (historical): Challenging reflects a global extremal problem with pivot-path constraints in every dimension; specialist impact concerns sharp constants on the orthogonal subclass.

Negative resolution - 2026-09-11

Solved. George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology, gives an exact order-eight counterexample in the [complete proof, Theorem and Sections 1-4](#). Let $\tilde{L} = L_8 + e_8 e_2^T$, changing only its (8,2) entry from -1 to 0 , and take its positive-diagonal QR factor $\tilde{Q}$. Under the stipulated first-available-row tie rule,

$$\rho_{\text{PP}}(\tilde{Q}) = \frac{5272}{63} > \sqrt{\frac{17948132}{2601}} = \rho_{\text{PP}}(Q_8).$$

The proof prints exact integer-column descriptions of both orthogonal matrices and verifies all active Schur-complement maxima. This disproves the universal extremizer equality. It does not determine the true orthogonal supremum or refute the separate asymptotic leading-constant conjecture. [Proof PDF](#) · [Standalone TeX](#).

A separate [Codex-agent mathematical review](#) returned **PASS**, with independently written rational checks of both QR conventions, all pivot ties, all 408 active entries and the positive squared growth gap. Substantial ChatGPT/Codex assistance is disclosed; this is automated-agent review, not external human peer review or formal certification. [Authorship, reproducible certificates, source checks and public-branch audit](#). The original ID, path, statement, historical ratings and earlier checks below are retained. Peca-Medlin retains credit for the conjecture and cited element-growth analysis.

Lean proof and verification evidence — 2026-09-14

The complete original extremizer conjecture has a Lean-verified negative answer. The complete universal orthogonal extremizer equality is disproved at order eight. Four exports establish the actual positive-diagonal QR factors, first-available-row pivot paths, every Schur update along those paths and strict growth separation, and the original conjecture's negation using a genuine nonempty bounded supremum over all orthogonal inputs and admissible paths. The true supremum is not determined.

Immutable formal proof. The checked exports are `NLA.IE05.qr_certificates`, `NLA.IE05.pivot_certificates`, `NLA.IE05.growth_separation`, `NLA.IE05.counterexample`. Lean 4.33.1 uses pinned Mathlib `0df444a3` and LeanCert `621a43d7`; [complete pins and reproduction instructions](#) are retained. All four exported transitive axiom closures contain only *propext*, *Classical.choice* and *Quot.sound*.

Formalization: Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation, with OpenAI Codex assistance. George Stepaniants retains mathematical authorship. Two independent statement reviews preceded implementation, and two independent final proof reviews passed. Kernel-only LeanCert and [fresh isolated Linux Comparator and kernel verification](#) passed, including rejection controls. [Original artifact, exact checked inputs and independent operational audit](#) are retained. This is AI-assisted formal verification, not external human peer review or source-author endorsement.

Context and notation

All elimination in this problem is in exact arithmetic. Write $\|A\|_{\max} = \max_{ij} |a_{ij}|$. A pivoting path creates successive active Schur complements $S_1 = A, S_2, \dots, S_n$, with row/column permutations as appropriate. Its element-growth factor is

$$\rho(A) = \frac{\max_{1 \leq j \leq n} \|S_j\|_{\max}}{\|A\|_{\max}}.$$

Partial pivoting chooses a largest-magnitude entry in the active first column; complete pivoting chooses one anywhere in the active matrix. If ties occur, a universal statement includes every admissible tie choice; a supremum includes all admissible paths. These conventions remove implementation-dependent ambiguity. Matrices are nonsingular unless otherwise stated.

Problem statement

Let L_n be the real unit lower triangular matrix whose entries strictly below the diagonal are all -1 . Define Q_n by the unique QR factorization $L_n = Q_n R_n$ with positive diagonal in R_n . For Q_n , use partial pivoting with the first available row chosen in a tie. Is it true that, for every $n \geq 2$,

$$\sup_{Q \in O(n)} \rho_{\text{PP}}(Q) = \rho_{\text{PP}}(Q_n), \quad O(n) = \{Q \in \mathbb{R}^{n \times n} : Q^T Q = I\}?$$

The supremum on the left includes all admissible partial-pivoting paths. The candidate is fully specified by L_n , rather than by an approximate numerical optimizer. This asks for the sharp extremizer, beyond the established exponential order of orthogonal growth.

Reference

Peca-Medlin, *Growth factors of orthogonal matrices and local behavior of Gaussian elimination with partial and complete pivoting*, published in SIAM J. Matrix Anal. Appl. (2024), §3.2 and Appendix B. The paper conjectures this equality and establishes $\rho_{\text{PP}}(Q_n) = 2^{n-1}(1 + o(1))/\sqrt{3}$.

Earlier status check — 2026-09-08

Searches for *GEPP orthogonal conjecture 2026* and the exact paper title found no proof of the extremal equality. The 2026 butterfly paper still identifies orthogonal partial-pivoting growth as open. The August Shah–Urschel results concern different growth questions and pivot strategies; their exponential examples do not establish this exact supremum.

Audit update — 2026-09-10

Rechecked [Peca-Medlin's manuscript](#), §3.2 and Appendix B, and its [SIAM publication](#), SIAM J. Matrix Anal. Appl. 45 (2024), 1599–1620. The candidate extremizers remain supported by the stated construction and experiments, without a general optimality proof. Searches for subsequent orthogonal GEPP extremizer results found no resolution.